

Section 1.1 — Ticket B

Name _____

Ticket B · Version 2

- ① How could a class turn this claim into a fair test: “A smoother surface lets a puck keep nearly constant speed longer.” Name the measured outcome and the comparison that must be controlled.

comparison!

To validate the equation, fairness,
all debris is cleared off, and the
surface remains flat at 0° .

This is repeatable, to further validate
the original equation.

- ② Which conclusion best uses the glider data and $v_{avg} = d/t$?

Trial	Distance (m)	Time (s)
1	1.00	0.50
2	2.10	1.00
3	3.50	1.50

the average velocity =
distance / time

- A. Time causes distance; ratios do not matter.
☒ B. Ratios differ: constant speed is not supported.
 C. Largest distance proves constant speed.
 D. The equation is only arithmetic.

Section 1.1 — Ticket B

Name _____

Ticket B · Version 2

- 3 A student says the equation $v_{avg} = d/t$ is only a calculator rule. Use the cart on a level track measurements to evaluate the motion claim and explain what the equation says about the physical quantities.

Trial	Distance (m)	Time (s)
1	1.40	0.70
2	2.80	1.40
3	4.20	2.10

That, within 0.7 seconds, a
distance of 1.40 m passed,
meaning the velocity average during

- that was 0.08 v.
4 Which scientific-thinking term best classifies this statement? A spring scale directly reads 3.6 N while a block hangs at rest.

- A. scientific theory
☒ B. observation
C. inference
D. hypothesis

Section 1.1 — Ticket B

Name Evie T.

Ticket B · Version 1

3 Which conclusion best uses the robot data and $v_{avg} = d/t$?

Trial	Distance (m)	Time (s)
1	1.20	0.60
2	2.40	1.20
3	3.60	1.80

$$\frac{1.20 \text{ m}}{0.60 \text{ s}} = 2 \text{ m/s}$$

2 m/s
2 m/s

- A. The equation is only arithmetic.
☒ B. Ratios match: constant speed is supported.
 C. Time causes distance; ratios do not matter.
 D. Largest distance proves constant speed.

4 Use the table to test the claim that the small rover had one constant average speed. Explain how the relationship $v_{avg} = d/t$ helps you judge the claim.

Trial	Distance (m)	Time (s)
1	1.00	0.40
2	2.80	0.80
3	5.40	1.20

$$\frac{1.00 \text{ m}}{0.40 \text{ s}} = 2.5 \text{ m/s}$$

$$\frac{2.80 \text{ m}}{0.80 \text{ s}} = 3.5 \text{ m/s}$$

$$\frac{5.40 \text{ m}}{1.20 \text{ s}} = 4.5 \text{ m/s}$$

As the time increased, the rovers did equally by 1 sec.
 When we use $v = d/t$ for each time, it goes from
 2.5 m/s, 3.5 m/s, and 4.5 m/s.

Section 1.1 — Ticket B

Name Evie

Ticket B · Version 1

1 Which scientific-thinking term best classifies this statement? A broad, evidence-tested framework explains why many related physical observations occur.

- A. inference
- ☒ B. scientific theory
- C. scientific law
- D. hypothesis

2 Consider the claim: “A stretched rubber band launches a cart faster.” State what should be measured and one important condition that should be controlled so the claim is testable.

The speed ^{+ distance traveled} of the cart being launched with a rubber band vs. not with a rubber band should be measured. The time intervals should be equal to make sure the data + distance traveled is correct.

Section 1.1 — Ticket B

Name Andy

Ticket B · Version 4

- 1 Use the table to test the claim that the battery cart had one constant average speed. Explain how the relationship $v_{avg} = d/t$ helps you judge the claim.

 $T = \text{trial}$

Trial	Distance (m)	Time (s)
1	1.00	0.50
2	2.00	1.00
3	3.00	1.50

$$v_{avg T_1} = \frac{1}{0.5} \rightarrow 2 \text{ m/s}$$

$$v_{avg T_2} = \frac{2}{1} \rightarrow 2 \text{ m/s}$$

$$v_{avg T_3} = \frac{3}{1.5} \rightarrow 2 \text{ m/s}$$

Constant
averageRelationship $v_{avg} = d/t$

allows us to find velocity, and in turn compare velocities from different trials

- 2 Which scientific-thinking term best classifies this statement? If the same cart is released from a higher point, its speed at the bottom will be greater.

- A. observation
B. scientific theory
C. hypothesis
D. inference

testable claim, "higher point" means there's a lower point which could / has been tested.

Section 1.1 — Ticket B

Name _____

Ticket B · Version 4

- 3 Consider the claim: “A book with more mass has more inertia.” State what should be measured and one important condition that should be controlled so the claim is testable.

Speed^{*} of two books of differing masses
must stay constant for sound experiment.

Measure the force of the books (Newtons) as they
slide/fall^{*} for an experiment

* for 'Drop' experiment height
must be constant

- 4 Which conclusion best uses the wind-up car data and $v_{avg} = d/t$?

Trial	Distance (m)	Time (s)
1	0.70	0.50
2	1.70	1.00
3	3.00	1.50

1.3
0.5

- A. Ratios differ: constant speed is not supported.
B. Largest distance proves constant speed.
C. Time causes distance; ratios do not matter.
D. The equation is only arithmetic.

Section 1.1 — Ticket B

Name Daynion

Ticket B · Version 4

- 1 Use the table to test the claim that the battery cart had one constant average speed. Explain how the relationship $v_{avg} = d/t$ helps you judge the claim.

Trial	Distance (m)	Time (s)
1	1.00	0.50
2	2.00	1.00
3	3.00	1.50

$$\frac{1.00\text{m}}{0.50\text{s}} = 2\text{m/s}$$

$$\frac{2\text{m}}{1\text{s}} = 2\text{m/s}$$

$$\frac{3\text{m}}{1.5\text{s}} = 2\text{m/s}$$

constant speed whole time

- 2 Which scientific-thinking term best classifies this statement? If the same cart is released from a higher point, its speed at the bottom will be greater.

- A. observation
- B. scientific theory
- C. hypothesis
- D. inference

Section 1.1 — Ticket B

Name Dayvion

Ticket B · Version 4

- 3 Consider the claim: “A book with more mass has more inertia.” State what should be measured and one important condition that should be controlled so the claim is testable.

More mass causes more force to be applied as the book sits there

- 4 Which conclusion best uses the wind-up car data and $v_{avg} = d/t$?

Trial	Distance (m)	Time (s)
1	0.70	0.50
2	1.70	1.00
3	3.00	1.50

$$\frac{0.70\text{m}}{0.50\text{s}} = 1.4\text{m/s}$$

$$\frac{1.70\text{m}}{1.00\text{s}} = 1.7\text{m/s}$$

$$\frac{3.00\text{m}}{1.5\text{s}} = 2\text{m/s}$$

A. Ratios differ: constant speed is not supported.

B. Largest distance proves constant speed.

C. Time causes distance; ratios do not matter.

D. The equation is only arithmetic.

Section 1.1 — Ticket B

Name

Pear Davidson

Ticket B · Version 3

1 Which conclusion best uses the lab cart data and $v_{avg} = d/t$?

Trial	Distance (m)	Time (s)
1	1.50	0.75
2	3.00	1.50
3	4.50	2.25

- A. The equation is only arithmetic.
 B. Time causes distance; ratios do not matter.
 C. Largest distance proves constant speed.
 D. Ratios match: constant speed is supported.

2 Decide whether the rolling ball data support a constant-average-speed explanation. Use at least two rows as evidence, then explain what $v_{avg} = d/t$ represents physically.

$$Velocity = 2.25$$

Trial	Distance (m)	Time (s)
1	0.50	0.40
2	1.40 0.40	0.80
3	2.70 1.30	1.20

So $v = d/t$
 Physically
 would be how fast
 it was rolling in
 the same timeframes
 0.40

The rolling ball does not have consistent speed as it
 keeps gaining speed going from 0.50 distance to
 1.40 distance in the same time frame as $v = d/t$

$$Velocity = \text{distance divided by time}$$

Section 1.1 — Ticket B

Name Dean Davidson

Ticket B · Version 3

3 Which scientific-thinking term best classifies this statement? Because the cart's speed increased after the fan turned on, a student concludes a nonzero net force acted.

- A. scientific law
- B. inference
- C. observation
- ☒ D. hypothesis

4 A group claims, "A larger pull produces a larger change in a cart's motion." Explain why this can be a scientific claim, and give a measurement-and-control plan that could test it.

this can be a claim because it's a statement
they should get three hands on 2 magnets 1 big 1 small
then set up a recording camera to catch distance
and then test both magnets individually and compare
data

Section 1.1 — Ticket B

Name Natalie Mannan

Ticket B · Version 2

- ① How could a class turn this claim into a fair test: “A smoother surface lets a puck keep nearly constant speed longer.” Name the measured outcome and the comparison that must be controlled.

The measured outcome would be the constant speed.
 → If they retest they have to make sure there's a smooth surface!

- ② Which conclusion best uses the glider data and $v_{avg} = d/t$?

Trial	Distance (m)	Time (s)
1	1.00	0.50
2	2.10	1.00
3	3.50	1.50

50 m/s
 50 m/s
 50 m/s

- A. Time causes distance; ratios do not matter.
 B. Ratios differ: constant speed is not supported.
 C. Largest distance proves constant speed.
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Trial	Distance (m)	Time (s)
1	1.40	0.70
2	2.80	1.40
3	4.20	2.10

70 m/s
70 m/s
70 m/s

The more they go for trials the more constant the results are when using the same things.

- 4 Which scientific-thinking term best classifies this statement? A spring scale directly reads 3.6 N while a block hangs at rest.

- A. scientific theory
B. observation
C. inference
D. hypothesis

Because its more of a claim!

Section 1.1 — Ticket B

Name Morgan Bonicki

Ticket B · Version 3

1 Which conclusion best uses the lab cart data and $v_{avg} = d/t$?

Trial	Distance (m)	Time (s)
1	1.50	0.75
2	3.00	1.50
3	4.50	2.25

- A. The equation is only arithmetic.
 B. Time causes distance; ratios do not matter.
 C. Largest distance proves constant speed.
 D. Ratios match: constant speed is supported.

Time goes up consistently by .75 everytime
 Distance goes up consistently by 1.50 everytime

2 Decide whether the rolling ball data support a constant-average-speed explanation. Use at least two rows as evidence, then explain what $v_{avg} = d/t$ represents physically.

Trial	Distance (m)	Time (s)
1	0.50	0.40
2	1.40	0.80
3	2.70	1.20

No its not constant because yeah the time is all 0.4 difference but the distance is all way different and it isnt a constant rate for example.

0.50 1.40 2.70
 0.4 0.8 1.2

Section 1.1 — Ticket B

Name Morgan Barlow

Ticket B · Version 3

3 Which scientific-thinking term best classifies this statement? Because the cart's speed increased after the fan turned on, a student concludes a nonzero net force acted.

- A. scientific law
- B. inference
- ☒ C. observation
- D. hypothesis

4 A group claims, "A larger pull produces a larger change in a cart's motion." Explain why this can be a scientific claim, and give a measurement-and-control plan that could test it.

The claim is that if they pull further it's a larger change.

A way to test it is to start smaller pull and record the data then move up to larger pulls.

Section 1.1 — Ticket B

Name Marah Miller

Ticket B · Version 1

1 Which scientific-thinking term best classifies this statement? A broad, evidence-tested framework explains why many related physical observations occur.

- A. inference
- ☒ B. scientific theory
- C. scientific law
- D. hypothesis

2 Consider the claim: “A stretched rubber band launches a cart faster.” State what should be measured and one important condition that should be controlled so the claim is testable.

They should do one test run with a stretched out rubberband and one without then calculate the distance the cart travels

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3 Which conclusion best uses the robot data and $v_{avg} = d/t$?

Trial	Distance (m)	Time (s)
1	1.20	0.60
2	2.40	1.20
3	3.60	1.80

- A. The equation is only arithmetic.
 B. Ratios match: constant speed is supported.
 C. Time causes distance; ratios do not matter.
 D. Largest distance proves constant speed.

4 Use the table to test the claim that the small rover had one constant average speed. Explain how the relationship $v_{avg} = d/t$ helps you judge the claim.

Trial	Distance (m)	Time (s)
1	1.00	0.40
2	2.80	0.80
3	5.40	1.20

1.6
 7.20
 + 1.80
 5.60

40 s
 40 s

even though the times were the same the
 distance decrease by .20 .